Part 1: Python SQLite Game Shop POS

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Time required: 60 minutes

- Follow all directions carefully and accurately.
- Think of the directions as minimum requirements.

SQL Tutorial

- https://www.w3schools.com/sql/sql_create_table.asp

SQLite with Python Tutorials

- SQLite Databases with Python - Full Course – FreeCodeCamp.org
- https://www.sqlitetutorial.net

SQLite Relational Database

SQLite is a relational database. We can create tables related by primary keys. We will design our databases using an ERD (Entity Relationship Diagram). www.lucidchart.com is free web-based diagram site used in these SQLite tutorials.

In this tutorial, we will create a games database with two related tables, game and publisher along with a bridge or transaction table.

Business Rules

Business rules establish how the entities interact and their relationship with each other. For example:

- A publisher can produce many games.
- A game can only have one publisher.
- A customer can purchase many games.
- A game can be purchased by many customers.

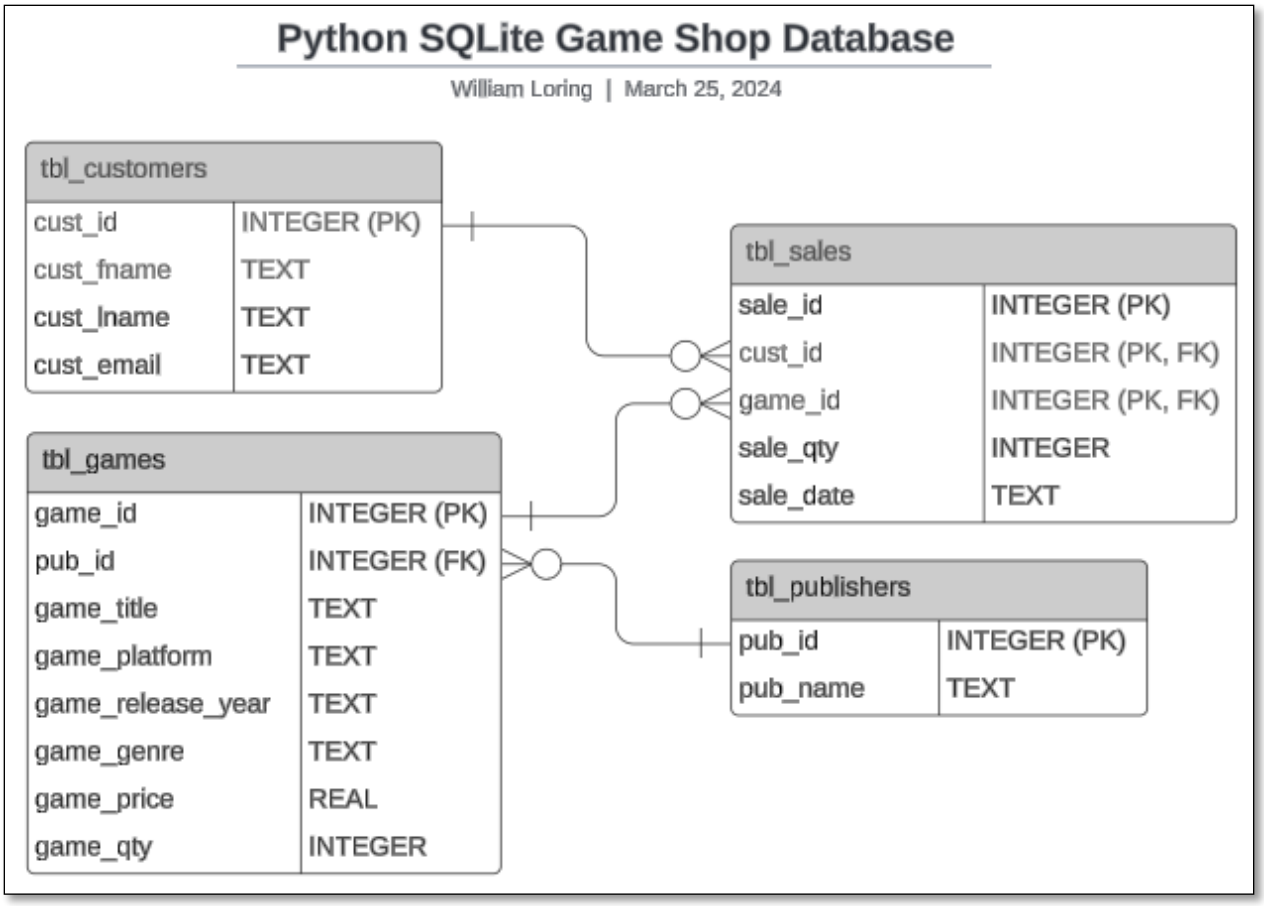
Game Shop Database ERD

An Entity Relationship Diagram, also known as ERD, is a diagram that displays the relationship of entities stored in a database. ER diagrams help to explain the logical structure of databases. ER diagrams are created based on three basic concepts: entities (tables), attributes (fields), and relationships.

An ERD of our movies related database is based on the following business rules:

- A publisher can produce many games.
- A game can only have one publisher.
- A customer can purchase many games.
- A game can be purchased by many customers.

This is a crow’s foot diagram illustrating a one to many relationship between the entities.

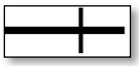


Reference: https://vertabelo.com/blog/crow-s-foot-notation/

PK (Primary Key): A primary key is a column or a set of columns in a table whose values uniquely identify a row in the table. A primary key typically has no meaning other than to uniquely identify each record.

FK (Foreign Key): A foreign key is a column or a set of columns in a table whose values correspond to the values of the primary key in another table.

Crows foot notation indicates the record relationship between the tables.

	One or many
	One

Assignment 1: Draw ERD

1. Take your ERD from your previous database project.
2. Draw a Crow's Foot ERD for a relational table version of your project similar to the example.
3. Create a Word Documentation file to store a screenshot of your ERD and your other planning items.

Assignment 2: Data Dictionary

Let's create your database and tables. Fill in your names in your planning document.

1. Create a Python file named:
2. Database file name:
3. Tables (Entities):
4. Fields (Attributes)

This is the structure of our sample database. This is called a data dictionary.

Table Name	Field Name	Field Format	Description
tbl_publishers	pub_id	INTEGER – PK	Game publisher unique identifier
tbl_publishers	pub_name	TEXT	Game publisher name
tbl_games	game_id	INTEGER – PK	Game unique identifier
tbl_games	pub_id	INTEGER – FK	Foreign key referencing the publisher associated with the game
tbl_games	game_title	TEXT	Game title
tbl_games	game_platform	TEXT	Game platform
tbl_games	game_release_year	INTEGER	Game release year
tbl_games	game_genre	TEXT	Game genre
tbl_games	game_price	REAL	Price of game
tbl_games	game_qty	INTEGER	Quantity of games on hand
tbl_sales	sale_id	INTEGER (PK)	Sale unique identifier
tbl_sales	sale_qty	INTEGER	Sale quantity
tbl_sales	sale_date	TEXT	Date of sale

tbl_sales	cust_id	(PK, FK)	Composite key for sales and customers
tbl_sales	game_id	(PK, FK)	Composite key for sales and games
tbl_customers	cust_id	INTEGER - PK	Customer unique identifier
tbl_customers	cust_f_name	TEXT	First name
tbl_customers	cust_l_name	TEXT	Last name
tbl_customers	cust_email	TEXT	Email Address

Assignment:

1. Update your data dictionary for your database in your planning document.
2. Add a Customer Email address.

Assignment Submission

1. Attach the program files.
2. Attach screenshots showing the successful operation of the program.
3. Submit in Blackboard.